

Pseudocode

Given Input

City_distances[] array

Gas_station[] array

Mpg int

Need # of cities, calculate using loop

num_cities

For x in city:

 Num_cities += 1

For main section need two loops to calculate algorithm

Inside loop starts at a city, then travels as far as possible

This is done via adding value in gas station array * mpg int and adding it to fuel

Then subtract # of miles to next city by fuel - city_distance to next city

Should only loop a number of times until it fails to travel to another city

Outside loop needs to change the starting city and reset the inside loops variables

Should only loop a number of times # to num cities max

Something along these lines

Loop for # of elements in city_distances or gas_station

{

 Starting_Pos = 0

 Current_Fuel = 0

 Cities_travelled = 0

 Position = starting

```

    {
        Fuel = Fuel + (Gas_Station[Position]*mpg)
        Fuel = Fuel - (City_Distances[Position])
        Position = Position + 1
        //Used to loop when you reach the end
        If Position > # of elements in City_distances
            Position = 0
        //Used before the loop check to preemptively increase the
        distance traveled
        If Fuel >= 0,
            Cites_travelled +1
        Stop_loop = Stop_loop + 1
    }
    Starting_pos + 1
}

```

Big Oh Notation Using Step Count:

To initialize the project takes 5 steps

$O(5)$

First loop happens upwards of n times

$O(n+5)$

Outside loop has 5 step and occurs the same number of n times. Because its a nested loop, we will eventually multiply the two loops together

$O(n+5+(5n*x))$

Inside loop has upwards of 7 steps and can occur upwards of n times as well so we can fill the x with $7n$

$O(n+5+(5n(7n)))$

Simplifying the expression gives us

$O(n+5+(35n^2))=(35n^2+n+5)$

Simplifying it further to get the time complexity gives us a time complexity of

$O(n^2)$